Cuatro cantidades que suelen confundirse

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$$\mathcal{L} = \frac{1}{2} m\dot{x}^2 e^{\alpha t/m} \qquad \alpha > 0$$

a)
$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} e^{\alpha t/m}$$

b) $m\dot{x}$

c) $h = \dot{x}\frac{\partial \mathcal{L}}{\partial \dot{x}} - \mathcal{L}$

$$h = \dot{x} m\dot{x} e^{\alpha t/m} - \mathcal{L}$$

$$h = \dot{x}^2 m e^{\alpha t/m} - \frac{1}{2} m\dot{x}^2 e^{\alpha t/m}$$

$$h = \frac{1}{2} m\dot{x}^2 e^{\alpha t/m}$$

$$\frac{d}{dt}\dot{x}^2 \cdot e^{\alpha t/m} = 2\dot{x}\cdot\ddot{x} e^{\alpha t/m} + \dot{x}^2 \frac{\alpha}{m} e^{\alpha t/m}$$

$$\frac{dh}{dt} = \frac{1}{2} m e^{\alpha t/m}\left(2\ddot{x} + \frac{\alpha \dot{x}}{m}\right)$$

$$\frac{dh}{dt} = e^{\alpha t/m}\left(m\ddot{x} + \frac{\alpha}{2}\dot{x}\right)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j}\right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \qquad \frac{\partial \mathcal{L}}{\partial x} = m\dot{x} e^{\alpha t/m} = p_x = cste \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = 0$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = m\left(\ddot{x} e^{\alpha t/m} + \dot{x}\frac{\alpha}{m} e^{\alpha t/m}\right)$$

$$= e^{\alpha t/m}\left(m\ddot{x} + \alpha\dot{x}\right)$$

$$e^{\alpha t/m}\left(m\ddot{x} + \alpha\dot{x}\right) = 0 \qquad e^{\alpha t/m} > 0$$

$$m\ddot{x} + \alpha\dot{x} = 0$$

$$\ddot{x} = -\frac{\alpha \dot{x}}{m}$$

$$\frac{dh}{dt} = e^{\alpha t/m}\dot{x}\left(\alpha\dot{x} + \frac{\alpha}{2}\dot{x}\right)$$

$$\frac{dh}{dt} = e^{\alpha t/m}\dot{x}\left(-\frac{\alpha}{2}\dot{x}\right)$$

$$= e^{\alpha t/m}\dot{x}^2\left(-\frac{\alpha}{2}\right)$$

$$\frac{dh}{dt} = -\frac{\alpha}{2} e^{\alpha t/m}\dot{x}^2 \qquad \dot{x}^2 > 0,\ \alpha > 0,\ e^{\alpha t/m} > 0$$

$$\hookrightarrow \frac{dh}{dt} < 0$$

d) $T = \frac{1}{2} m\dot{x}^2$ $\qquad \ddot{x} = -\frac{\alpha}{m}\dot{x}$

$$\frac{dT}{dt} = m\dot{x}\ddot{x}$$

$$\frac{dT}{dt} = m\dot{x}\left(-\frac{\alpha}{m}\dot{x}\right)$$

$$\frac{dT}{dh} = -\alpha\dot{x}^2 \qquad \dot{x}^2 > 0 \Rightarrow \frac{dT}{dh} < 0$$

e)

$$\frac{dp_x}{dt} = 0 \rightarrow \text{se conserva}$$

momento mecánico

$$\frac{d(m\dot{x})}{dt} = m\ddot{x} = -\alpha\dot{x} \neq 0 \quad \text{no se conserva}$$

$$\frac{dT}{dt} = -\alpha\dot{x}^2 < 0 \quad \text{no se conserva (Disminuye)}$$

f) $t \rightarrow t + \varepsilon$

Esto es una simetría si

$$\mathcal{L}(t) = \mathcal{L}(t+\varepsilon)$$

$$e^{\alpha t/m} = e^{\alpha(t+\varepsilon)/m} \; ? \rightarrow \text{No, difieren por } e^{\alpha\varepsilon/m}$$

$$\frac{\partial \mathcal{L}}{\partial t} = 0 \Rightarrow \text{La energía se conserva}$$

$$\frac{\partial \mathcal{L}}{\partial t} = \frac{1}{2} m\dot{x}^2 \cdot \frac{\alpha}{m} e^{\alpha t/m} = \frac{\alpha}{2}\dot{x}^2 e^{\alpha t/m}$$

$$\frac{\partial \mathcal{L}}{\partial t} \neq 0$$

Por lo tanto la energía no se conserva.

g) $h = \dot{x}\frac{\partial \mathcal{L}}{\partial \dot{x}} - \mathcal{L}$

$$T = \frac{1}{2} m\dot{x}^2$$

$$\mathcal{L} = T(\dot{x}) - V(x)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{\partial T}{\partial \dot{x}}$$

$$h = \dot{x}\frac{\partial \mathcal{L}}{\partial \dot{x}} - \mathcal{L}$$

$$= \dot{x}\frac{\partial T}{\partial \dot{x}} - (T - V)$$

$$= \dot{x}\frac{\partial T}{\partial \dot{x}} - T + V$$

$$\dot{x}\frac{\partial T}{\partial \dot{x}} = 2T$$

$$h = 2T - T + V = T + V$$

Condiciones

1. $\mathcal{L} = T(\dot{x}) - V(x)$
2. Energía cinética en función solo de la velocidad
3. Potencial solo en términos de la posición
4. No hay dependencia explícita del tiempo

No se cumple en nuestro caso porque $e^{\alpha t/m}$ implica que rompe (4)